

| Assessment | Test | Domain  | Skill            | Difficulty                                          |
|------------|------|---------|------------------|-----------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Medium <div><div></div><div></div><div></div></div> |

Question

In the linear function  $h$ ,  $h(28) = 15$  and  $h(26) = 22$ . Which equation defines  $h$ ?

Answer

- A.  $h(x) = -\frac{2}{7}x + 23$
- B.  $h(x) = -\frac{2}{7}x + 113$
- C.  $h(x) = -\frac{7}{2}x + 23$
- D.  $h(x) = -\frac{7}{2}x + 113$

Correct Answer: D

Rationale

Choice D is correct. An equation defining  $h$  can be written in the form  $y = mx + b$ , where  $y = h(x)$ ,  $m$  represents the slope of the graph of  $y = h(x)$  in the  $xy$ -plane, and  $b$  represents the  $y$ -coordinate of the  $y$ -intercept of the graph. It's given that  $h(28) = 15$  and  $h(26) = 22$ . It follows that the points  $(28, 15)$  and  $(26, 22)$  are on the graph of  $y = h(x)$  in the  $xy$ -plane. The slope can be found by using any two points,  $(x_1, y_1)$  and  $(x_2, y_2)$ , and the formula  $m = \frac{y_2 - y_1}{x_2 - x_1}$ . Substituting  $(28, 15)$  and  $(26, 22)$  for  $(x_1, y_1)$  and  $(x_2, y_2)$ , respectively, in the slope formula yields  $m = \frac{22 - 15}{26 - 28}$ , or  $m = -\frac{7}{2}$ . Substituting  $-\frac{7}{2}$  for  $m$  and  $(28, 15)$  for  $(x, y)$  in the equation  $y = mx + b$  yields  $15 = (-\frac{7}{2})(28) + b$ , or  $15 = -98 + b$ . Adding 98 to both sides of this equation yields  $113 = b$ . Substituting  $-\frac{7}{2}$  for  $m$  and 113 for  $b$  in the equation  $y = mx + b$  yields  $y = -\frac{7}{2}x + 113$ . Since  $y = h(x)$ , it follows that the equation that defines  $h$  is  $h(x) = -\frac{7}{2}x + 113$ .

Choice A is incorrect. For this function,  $h(26) = \frac{109}{7}$ , not  $h(26) = 22$ .

Choice B is incorrect. For this function,  $h(28) = 105$ , not  $h(28) = 15$ , and  $h(26) = \frac{739}{7}$ , not  $h(26) = 22$ .

Choice C is incorrect. For this function,  $h(28) = -75$ , not  $h(28) = 15$ , and  $h(26) = -68$ , not  $h(26) = 22$ .